

Thermal physics · Mock midterm test · Version B · Solutions

Total 44 marks: I = 8, II = 14, III = 14, IV = 8. Marks for each part are shown on the right. A mark in square brackets [1] is one point earned by the step it follows. If you used a wrong value consistently in later steps (error carried forward), you still earn the later method marks. Section IV is answered in Italian.

I · Multiple choice

[8]

#	Answer	Marks
1	D — The kettle has a high temperature and a lot of internal energy; heat is energy being transferred, not energy stored.	1
2	A — The two blocks are in thermal equilibrium with each other.	1
3	C — $-196\text{ }^{\circ}\text{C}$	1
4	B — There is a lowest possible temperature, 0 K.	1
5	A — Group 2's thermometer is more sensitive.	1
6	D — 6.0 mm	1
7	B — It increases by the same fraction as the outer diameter.	1
8	C — 0.15 %	1

II · Paper 6 on the thermistor data

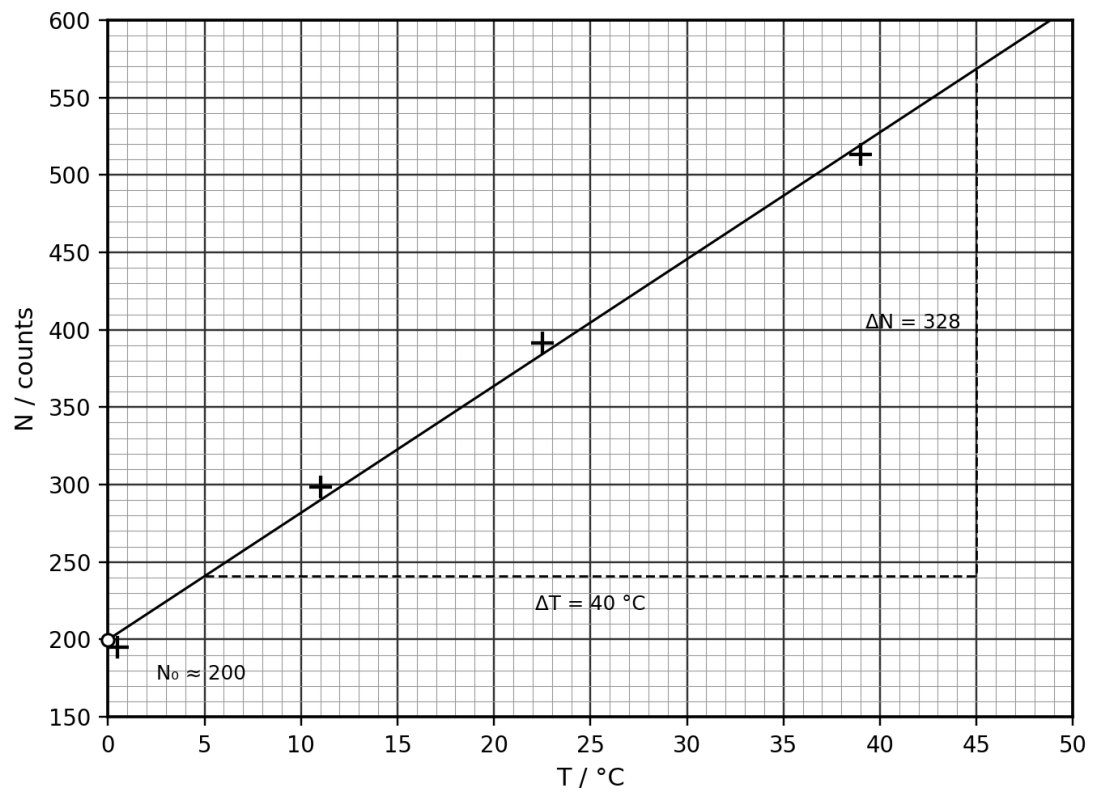
[14]

Data with the anomalous reading in square brackets and the correct means (the mean for bath 3 leaves out 426):

Bath	T / °C	N ₁	N ₂	N ₃	N ₄	N ₅	mean N
1 (ice water)	0.5	193	195	197	196	193	194.8
2 (cold water)	11.0	299	299	298	299	297	298.4
3 (room temperature)	22.5	390	[426]	392	391	393	391.5
4 (hot water)	39.0	512	513	512	513	515	513.0

Least-squares line through the four means: $m = 8.19 \text{ counts/}^\circ\text{C}$, $N_0 = 199.9$, so $T = (N - 199.9) / 8.19$. A line drawn by hand will give slightly different values (m within about $0.5 \text{ counts/}^\circ\text{C}$, N_0 within about 8 counts is fine). The test bath ($N = 429$) gives $T = 27.96 \text{ }^\circ\text{C}$. The reference thermometer reads $30.6 \text{ }^\circ\text{C}$: difference $2.64 \text{ }^\circ\text{C}$.

	Answer	Marks
(a)	Independent variable: temperature T of the bath (read on the reference thermometer). Dependent variable: thermistor reading N.	2
(b)	The reading 426 in bath 3 (N ₂), because it differs from the other four readings of that bath by about 35 counts, while those agree to within 3 counts.	1
(c)	Mean of the four remaining readings: 391.5.	1
(d)	Axes labelled "T / °C" (x) and "N / counts" (y). Four points plotted to within half a small square, each marked with a small cross (the centre of the cross is the point).	3
(e)	A single, straight, thin line drawn with a ruler, passing as close as possible to all the points, with points on both sides of it (not dot-to-dot, not forced through the first and last points).	1
(f)	Triangle drawn with a base of at least half the line (e.g. T = 5 to 45 °C); $m = \Delta N / \Delta T = 8.19 \text{ counts/}^\circ\text{C}$.	2
(g)	$N_0 \approx 200$ (read at T = 0 °C; fit: 199.9). Equation $T = (N - 199.9) / 8.19$.	1
(h)	$T = (429 - 199.9) / 8.19 \approx 28.0 \text{ }^\circ\text{C}$ (accepted: 26.5–29.5 °C).	1
(i)	Does NOT meet the requirement: $ 30.6 - 28.0 = 2.64 \text{ }^\circ\text{C} > 1 \text{ }^\circ\text{C}$.	1
(j)	One specific improvement, e.g.: take several readings in the test bath; use a reference thermometer with smaller divisions (0.1 °C resolution); stir the bath and wait for equilibrium before reading; use more calibration baths; use a second independent test bath.	1



Reference graph: points = means, least-squares line, gradient triangle from $T = 5$ to $T = 45$ $^{\circ}\text{C}$, intercept N_0 on the axis.

III · Structured questions

[14]

	Answer	Marks
1(a)	(i) $56\text{ }^{\circ}\text{C} = 329\text{ K}$; (ii) $4\text{ K} = -269\text{ }^{\circ}\text{C}$.	2
1(b)	Heat is not something a body has. Temperature is a property (state quantity) of the body; heat is energy transferred because of a temperature difference. What the body has is internal energy. [1] heat = energy in transfer because of ΔT (a process, not stored). [1] the body has a temperature / internal energy instead.	2
1(c)	0 K (absolute zero) is the lowest possible temperature; nothing can be cooled below it, so -5 K does not exist.	1
2(a)	$0\text{ }^{\circ}\text{C}$: melting point of pure ice. $100\text{ }^{\circ}\text{C}$: boiling point of pure water (at standard atmospheric pressure).	2
2(b)	The glass bulb cools down first and contracts, so the space inside the bulb decreases and pushes the liquid level up (1). Then the liquid cools and contracts more than the glass, so the column falls (1).	2
3(a)	$\Delta T = 35 - (-5) = 40\text{ }^{\circ}\text{C}$ [1]; $\Delta L = \alpha L_0 \Delta T = 17.0 \times 10^{-6} \times 5.0 \times 40$ [1] $= 3.4 \times 10^{-3}\text{ m} = 3.4\text{ mm}$ [1].	3
3(b)	$\beta = 2\alpha = 34.0 \times 10^{-6}\text{ }^{\circ}\text{C}^{-1}$ [1]; $\Delta A = \beta A_0 \Delta T = 34.0 \times 10^{-6} \times 0.60 \times 40 = 8.16 \times 10^{-4}\text{ m}^2$ (8.2 cm^2) [1].	2

IV · Problema — Boccola di alluminio in un compressore**[8]**

	Answer	Marks
(a)	Aumenta: il foro si dilata come se fosse fatto dello stesso materiale (come un disco pieno); tutte le lunghezze del pezzo scalano con la stessa frazione.	1
(b)	Diametro minimo del foro: $40,05 + 0,04 = 40,09$ mm [1]. $\Delta D = 0,09$ mm, quindi $\Delta T = \Delta D / (\alpha D) = 0,09 / (23,0 \times 10^{-6} \times 40,00) = 97,8$ °C [1]. $T = 20 + 97,8 = 117,8$ °C, minore del limite di 200 °C: compatibile [1].	3
(c)	Ogni lunghezza cresce di $\Delta D/D = 0,225$ %; l'area, che dipende dal quadrato del diametro, cresce circa del doppio: $\Delta A/A \approx 2 \alpha \Delta T = \beta \Delta T = 0,45$ % [1 per $\beta = 2\alpha$, 1 per il risultato]. Non serve conoscere il diametro ($\Delta A \approx 5,65$ mm ² per $D = 40,00$ mm).	2
(d)	Interferenza a 20 °C: 0,05 mm. A 150 °C: albero 40,1125 mm, foro 40,1196 mm, interferenza $-0,0071$ mm, cioè un gioco di 0,0071 mm [1]. Il collegamento NON resta serrato: l'alluminio ha α maggiore dell'acciaio, il foro cresce più dell'albero; l'interferenza si annulla già a 133,8 °C [1].	2

Dati utili e dati di disturbo. Servono: diametri, gioco richiesto, coefficienti α , limite del costruttore (per la verifica in (b)), temperatura di esercizio (per (d)). Non servono: massa, altezza o spessore del pezzo, potenza del forno, tempo di riscaldamento, pezzi prodotti, giri/min.